# Rapport Management - Forecast & Promotion Touristique

## Besoin metier

Identifier les pays et destinations a promouvoir en priorite, sous contrainte de budget et de risque externe, sans supposer que les donnees brutes sont parfaites.

## Problematique

La demande touristique future est estimee a partir d'un signal mensuel pays. La recommandation destination combine cette prevision avec qualite, sentiment, cout, meteo et historique campagne.

## Architecture

Le projet suit une architecture bronze/raw, processed et gold. Les fichiers bruts sont conserves intacts dans `data/raw`; les controles et transformations sont versionnables dans `src`.

## Nettoyage et gouvernance

Les pays sont harmonises, les destinations anonymisees sont mappees vers des noms metier, la colonne `destination_original` conserve la tracabilite source, les formats numeriques sont parses, les budgets non numeriques restent manquants, et les contradictions de campagne sont signalees plutot que masquees.

## GOLD DATA

La GOLD DATA est a granularite pays-destination. Les variables derivees principales sont `demand_growth`, `tourism_score`, `quality_score`, `sentiment_score`, `weather_penalty`, `campaign_efficiency`, `forecasted_demand` et `marketing_priority`.

## Choix IA

Le probleme est une prevision supervisee de serie temporelle mensuelle. Le split est temporel afin d'eviter une fuite d'information. Deux baselines sont comparees a Random Forest et Gradient Boosting.

## Resultats modeles

| Model | Approach type | MAE | RMSE | MAPE | R2 | Business interpretation | Status |
|:---|:---|---:|---:|---:|---:|:---|:---|

| Gradient Boosting + lag features | Supervised forecasting with lag features | 6.85324 | 8.67352 | 7.30196 | 0.855826 | Candidate model for country demand forecasting and marketing prioritization. | OK |

| Random Forest + lag features | Supervised forecasting with lag features | 7.69015 | 9.63958 | 8.15029 | 0.821921 | Candidate model for country demand forecasting and marketing prioritization. | OK |

| Baseline Persistence | Temporal baseline | 9.93929 | 12.1561 | 10.7011 | 0.716806 | Reference point used to prove whether advanced AI adds value. | OK |

| Baseline Moving Average | Temporal baseline | 13.2375 | 14.9413 | 14.3113 | 0.572171 | Reference point used to prove whether advanced AI adds value. | OK |

| SARIMAX | Statistical time-series model | 16.8858 | 20.3908 | 19.1977 | 0.203169 | Candidate model for country demand forecasting and marketing prioritization. | OK |

| Prophet | Additive time-series model | nan | nan | nan | nan | Model unavailable or no valid predictions. | Unavailable |

# Data Quality

Les controles couvrent valeurs manquantes, types, cardinalites et contradictions. Extrait:

| dataset | column | dtype | rows | missing | missing_rate | unique_values |

|:-------------|:---------------------|:---------------|-------:|----------:|---------------:|----------------:|

| destinations | country | object | 160 | 0 | 0 | 8 |

| destinations | destination | object | 160 | 0 | 0 | 160 |

| destinations | attractiveness | float64 | 160 | 0 | 0 | 123 |

| destinations | cost | int64 | 160 | 0 | 0 | 155 |

| destinations | rating | float64 | 160 | 0 | 0 | 103 |

| destinations | visitors | int64 | 160 | 0 | 0 | 160 |

| destinations | destination_original | object | 160 | 0 | 0 | 50 |

| market | country | object | 287 | 0 | 0 | 8 |

| market | month | datetime64[ns] | 287 | 0 | 0 | 36 |

| market | demand_index | float64 | 287 | 0 | 0 | 243 |

| reviews | country | object | 800 | 0 | 0 | 8 |

| reviews | destination | object | 800 | 0 | 0 | 183 |

# Recommandations

Top recommandations:

| country | destination | destination_original | rank_country | marketing_priority | forecasted_demand | quality_score | sentiment_score | weather_penalty | campaign_efficiency | allocated_budget | recommendation_reason |
|:----------|:--------------|:-----------------------|---------------:|---------------------:|--------------------:|----------------:|------------------:|------------------:|----------------------:|-------------------:|:-----------------------------------------------------------------------------|
| Morocco | Fès | City_39 | 1 | 0.65615 | 2.07459e+07 | 0.558633 | 1 | 0 | 0 | 12100 | Demand forecast=20745947, quality=0.56, sentiment=1.00, weather penalty=0.00 |
| Japan | Hiroshima | City_3 | 1 | 0.580756 | 1.66319e+07 | 0.672121 | 1 | 0.25 | 0 | 10700 | Demand forecast=16631943, quality=0.67, sentiment=1.00, weather penalty=0.25 |
| Spain | Madrid | City_10 | 1 | 0.554049 | 2.05248e+07 | 0.601535 | 0 | 0.25 | 2.93333 | 10300 | Demand forecast=20524842, quality=0.60, sentiment=0.00, weather penalty=0.25 |
| Portugal | Faro | City_43 | 1 | 0.540911 | 1.88256e+07 | 0.412457 | 0 | 0.25 | 0 | 10000 | Demand forecast=18825614, quality=0.41, sentiment=0.00, weather penalty=0.25 |
| USA | Las Vegas | City_17 | 1 | 0.538001 | 1.97061e+07 | 0.77496 | 0 | 0 | 0 | 10000 | Demand forecast=19706125, quality=0.77, sentiment=0.00, weather penalty=0.00 |
| Italy | Venice | City_37 | 1 | 0.53161 | 1.64617e+07 | 0.451757 | 1 | 0.25 | 0 | 9800 | Demand forecast=16461692, quality=0.45, sentiment=1.00, weather penalty=0.25 |
| Japan | Osaka | City_6 | 2 | 0.528362 | 1.63783e+07 | 0.51241 | 1 | 0.55 | 0 | 9800 | Demand forecast=16378337, quality=0.51, sentiment=1.00, weather penalty=0.55 |
| Japan | Kobe | City_31 | 3 | 0.522484 | 1.43191e+07 | 0.48746 | 1 | 0.25 | 0 | 9700 | Demand forecast=14319135, quality=0.49, sentiment=1.00, weather penalty=0.25 |
| Portugal | Coimbra | City_8 | 2 | 0.521188 | 1.37002e+07 | 0.868498 | 0.666667 | 0.25 | 0 | 9600 | Demand forecast=13700188, quality=0.87, sentiment=0.67, weather penalty=0.25 |
| Morocco | Tanger | City_14 | 2 | 0.515389 | 1.85371e+07 | 0.619178 | 0 | 0.25 | 0 | 9500 | Demand forecast=18537125, quality=0.62, sentiment=0.00, weather penalty=0.25 |

# Limites, biais et risques

Les destinations sont anonymisees, la cible officielle est un texte non tabulaire, les avis sociaux sont sujets a biais de representation, et la meteo est categorielle. Les scores doivent donc etre utilises comme aide a la decision, pas comme verite automatique.

# Pistes futures

Connecter des donnees calendaires reelles, ajouter elasticite prix, saisonnalite par destination, tests A/B campagne, et monitoring MLOps des performances.